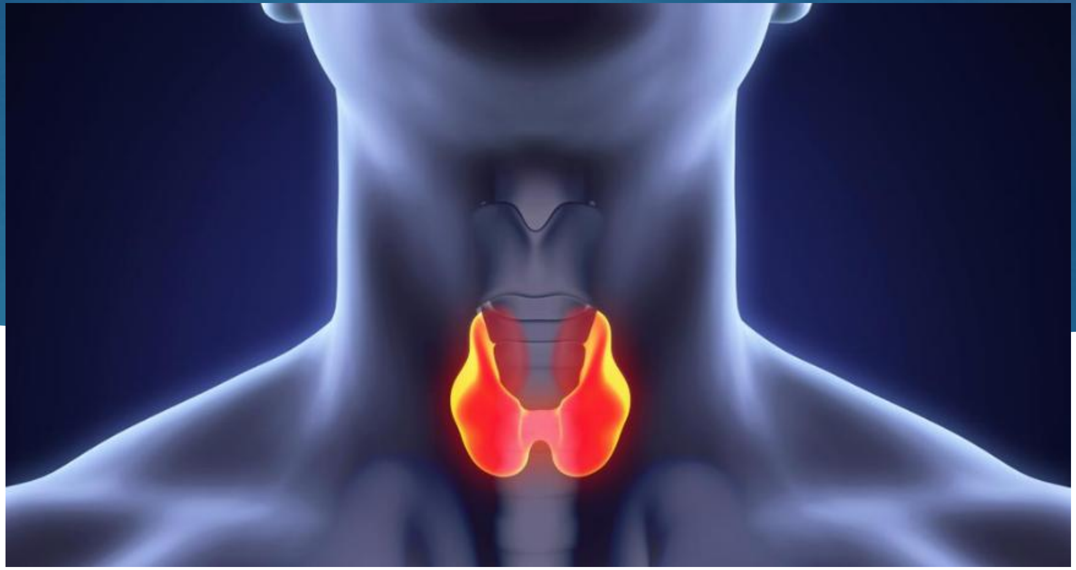
APPLYING ADVANCED REGRESSION MODELS AND MACHINE LEARNING FOR DIFFERENTIATED THYROID CANCER PATIENTS WITH RECURRENCES BY MARY SEMERDJIAN

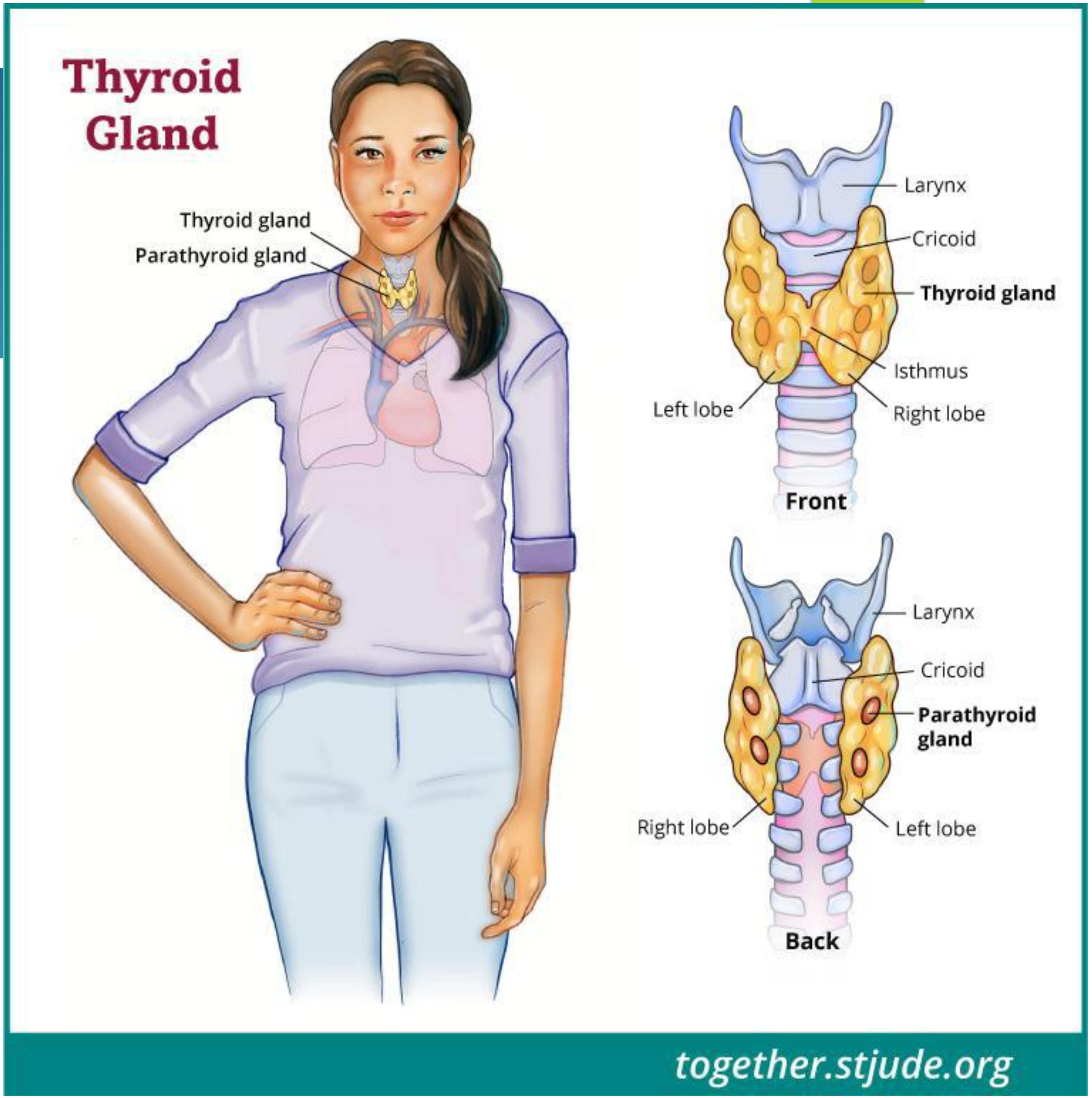


DTC Background

- Differentiated Thyroid Cancer (DTC) is a type of thyroid cancer in which the cancer cells look similar to normal thyroid cells when viewed under a microscope.

- Most differentiated thyroid cancers form in the follicular cells of the thyroid gland, that make hormones.

- Thyroid gland originates within differentiated thyroid cancer.



Dataset, Advanced Regression Models and Machine Learning

01 DTCR Dataset

- Provided by the University of California, Irvine, for health and medicine

02 Generalized Linear Model

- Represents a relation between the response variable y and the predictor variables

03 Binary Logistic Regression Model

- Utilizes a binary outcome to predict 1 and 0 and “yes” and “no”

04 Probit Model

- Uses a binary response variable y and predictors

05 Complementary Log-Log Model

- Make use of a binary response variable y and predictors

06 Random Forest

- For classification, prediction, and regression tasks and uses ensemble learning methods

Differentiated Thyroid Cancer Recurrence Dataset

```
spc_tbl_ [383 × 17] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ age                : num [1:383] 27 34 30 62 62 52 41 46 51 40 ...
 $ gender             : chr [1:383] "F" "F" "F" "F" ...
 $ smoking            : chr [1:383] "No" "No" "No" "No" ...
 $ hxsmoking          : chr [1:383] "No" "Yes" "No" "No" ...
 $ hxradiotherapy     : chr [1:383] "No" "No" "No" "No" ...
 $ thyroidfunction    : chr [1:383] "Euthyroid" "Euthyroid" "Euthyroid" "Euthyroid" ...
 $ physicalexamination: chr [1:383] "Single nodular goiter-left" "Multinodular goiter" "Single nodular goiter-right" ...
 $ adenopathy         : chr [1:383] "No" "No" "No" "No" ...
 $ pathology          : chr [1:383] "Micropapillary" "Micropapillary" "Micropapillary" "Micropapillary" ...
 $ focality           : chr [1:383] "Uni-Focal" "Uni-Focal" "Uni-Focal" "Uni-Focal" ...
 $ risk               : chr [1:383] "Low" "Low" "Low" "Low" ...
 $ t                  : chr [1:383] "T1a" "T1a" "T1a" "T1a" ...
 $ n                  : chr [1:383] "N0" "N0" "N0" "N0" ...
 $ m                  : chr [1:383] "M0" "M0" "M0" "M0" ...
 $ stage              : chr [1:383] "I" "I" "I" "I" ...
 $ response           : chr [1:383] "Indeterminate" "Excellent" "Excellent" "Excellent" ...
 $ recurred           : chr [1:383] "No" "No" "No" "No" ...
```

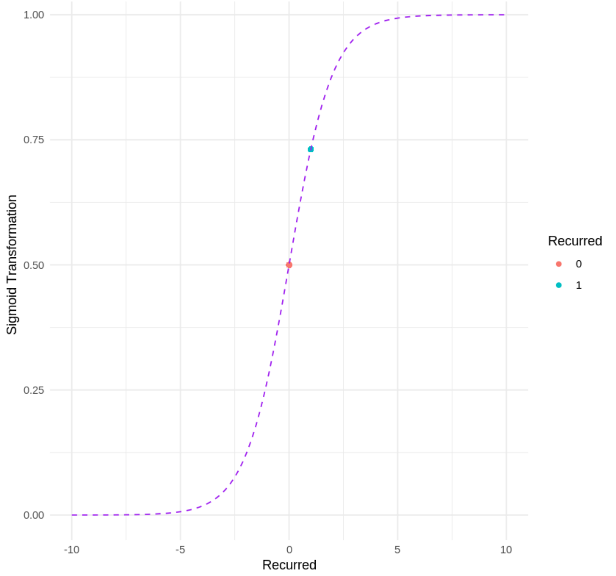
```
      age            gender    smoking    hxsmoking  hxradiotherapy
 Min.   :15.00   F:293    No :315    No :336    No :357
 1st Qu.:30.00   M: 71    Yes: 49    Yes: 28    Yes:  7
 Median :38.00
 Mean   :41.25
 3rd Qu.:52.00
 Max.   :82.00

                     thyroidfunction                  physicalexamination
 Clinical Hyperthyroidism  : 20    Diffuse goiter             :  7
 Clinical Hypothyroidism   : 12    Multinodular goiter        :135
 Euthyroid                 :313    Normal                     :  7
 Subclinical Hyperthyroidism:  5   Single nodular goiter-left : 88
 Subclinical Hypothyroidism : 14   Single nodular goiter-right:127

     adenopathy          pathology          focality            risk
 Bilateral: 32    Follicular    : 28   Multi-Focal:136   High        : 32
 Extensive:  7    Hurthel cell  : 20   Uni-Focal  :228   Intermediate:102
 Left     : 17    Micropapillary: 45                     Low         :230
 No       :258    Papillary     :271
 Posterior:  2
 Right    : 48

    t          n          m         stage                    response       recurred
 T1a: 46   N0 :249   M0:346   I   :314   Biochemical Incomplete: 23   No :256
 T1b: 40   N1a: 22   M1: 18   II  : 32   Excellent             :189   Yes:108
 T2 :138   N1b: 93            III:  4   Indeterminate         : 61
 T3a: 96                      IVA:  3   Structural Incomplete : 91
 T3b: 16                      IVB: 11
 T4a: 20
 T4b:  8
```

```
tibble [364 × 17] (S3: tbl_df/tbl/data.frame)
 $ age                : num [1:364] 27 34 30 62 62 52 41 46 51 40 ...
 $ gender             : Factor w/ 2 levels "F","M": 1 1 1 1 1 2 1 1 1 1 ...
 $ smoking            : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 2 1 1 1 1 ...
 $ hxsmoking          : Factor w/ 2 levels "No","Yes": 1 2 1 1 1 1 2 1 1 1 ...
 $ hxradiotherapy     : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 1 1 1 1 ...
 $ thyroidfunction    : Factor w/ 5 levels "Clinical Hyperthyroidism",..: 3 3 3 3 3 3 1 3 3 3 ...
 $ physicalexamination: Factor w/ 5 levels "Diffuse goiter",..: 4 2 5 5 2 2 5 5 5 5 ...
 $ adenopathy         : Factor w/ 6 levels "Bilateral","Extensive",..: 4 4 4 4 4 4 4 4 4 4 ...
 $ pathology          : Factor w/ 4 levels "Follicular","Hurthel cell",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ focality           : Factor w/ 2 levels "Multi-Focal",..: 2 2 2 2 1 1 2 2 2 2 ...
 $ risk               : Factor w/ 3 levels "High","Intermediate",..: 3 3 3 3 3 3 3 3 3 3 ...
 $ t                  : Factor w/ 7 levels "T1a","T1b","T2",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ n                  : Factor w/ 3 levels "N0","N1a","N1b": 1 1 1 1 1 1 1 1 1 1 ...
 $ m                  : Factor w/ 2 levels "M0","M1": 1 1 1 1 1 1 1 1 1 1 ...
 $ stage              : Factor w/ 5 levels "I","II","III",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ response           : Factor w/ 4 levels "Biochemical Incomplete",..: 3 2 2 2 2 3 2 2 2 2 ...
 $ recurred           : num [1:364] 0 0 0 0 0 0 0 0 0 0 ...
```



- Sigmoid Function Applied to the Differentiated Thyroid Cancer Recurrence data with Sigmoid Curve

Generalized Linear Model

```
glm(formula = recurred ~ age + gender.rel + hxsmoking.rel + thyroidfunction.rel +
    t.rel + n.rel + m.rel + stage.rel, family = gaussian(link = identity),
    data = differentiated_thyroid_cancer_recurrence.data)

Coefficients:
                                            Estimate Std. Error t value
(Intercept)                                 0.366567   0.288763   1.269
age                                         0.001850   0.001384   1.337
gender.relF                                -0.141126   0.044220  -3.191
hxsmoking.relYes                            0.040116   0.066156   0.606
thyroidfunction.relClinical Hyperthyroidism -0.045039  0.156943  -0.287
thyroidfunction.relClinical Hypothyroidism  0.076381   0.166945   0.458
thyroidfunction.relEuthyroid                0.076396   0.142237   0.537
thyroidfunction.relSubclinical Hypothyroidism -0.152601 0.167460 -0.911
t.relT1b                                    0.084858   0.068072   1.247
t.relT2                                     0.034236   0.053819   0.636
t.relT3a                                    0.181732   0.059046   3.078
t.relT3b                                    0.482372   0.098735   4.886
t.relT4a                                    0.406101   0.102551   3.960
t.relT4b                                    0.056941   0.161002   0.354
n.relN0                                    -0.222992   0.073903  -3.017
n.relN1b                                    0.241382   0.076831   3.142
m.relM0                                    -0.086819   0.118926  -0.730
stage.relI                                 -0.064402   0.189350  -0.340
stage.relII                                 0.104550   0.185950   0.562
stage.relIVA                                0.414164   0.257835   1.606
stage.relIVB                                0.192349   0.223712   0.860
```

```
                                             Pr(>|t|)
(Intercept)                                  0.20514
age                                          0.18223
gender.relF                                  0.00155 **
hxsmoking.relYes                             0.54465
thyroidfunction.relClinical Hyperthyroidism  0.77430
thyroidfunction.relClinical Hypothyroidism   0.64759
thyroidfunction.relEuthyroid                 0.59154
thyroidfunction.relSubclinical Hypothyroidism 0.36280
t.relT1b                                     0.21340
t.relT2                                      0.52512
t.relT3a                                     0.00225 **
t.relT3b                                     1.59e-06 ***
t.relT4a                                     9.12e-05 ***
t.relT4b                                     0.72381
n.relN0                                      0.00274 **
n.relN1b                                     0.00183 **
m.relM0                                      0.46588
stage.relI                                   0.73397
stage.relII                                  0.57431
stage.relIVA                                 0.10912
stage.relIVB                                 0.39050
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for gaussian family taken to be 0.09456176)
```

```
    Null deviance: 75.956  on 363  degrees of freedom
Residual deviance: 32.435  on 343  degrees of freedom
AIC: 196.86

Number of Fisher Scoring iterations: 2
0.307508964036492
```

- Gender, T3a, T3b, T4a, N0, N1b, however, are statistically significant predictors of differentiated thyroid cancer recurrence score at the 5% level of significance.

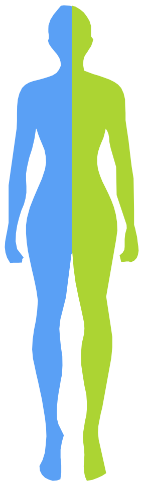
```
[58] print(predict(fitted.model, data.frame(age=55, gender.rel="F", hxsmoking.rel="Yes", n.rel="N1b",
     t.rel="T1b", m.rel="M0",thyroidfunction.rel="Subclinical Hypothyroidism", stage.rel="II")))

             1
     0.5586818
```

```
[59] pred = 0.3665+0.0018*55-0.1411+0.0401-0.1526+0.0849+0.2414-0.0868+0.1046
     print(pred)

     [1] 0.556
```

The predicted history of smoking score for a recurrence in patients with differentiated thyroid cancer where the patient is a 55-year-old female patient recurrence with different thyroid cancer who has a history of smoking, with N1b, T1b, M0, a subclinical hypothyroidism, and is in stage II is 55.6%.



Binary Logistic Regression Model

```
glm(formula = recurred ~ age + gender.rel + hxsmoking.rel + n.rel +
    response.rel, family = binomial(link = logit), data = differentiated_thyroid_cancer_recurrence.data)

Coefficients:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                       1.32041    1.68996   0.781 0.434613
age                               0.04856    0.02183   2.225 0.026078 *
gender.relF                      -0.92738    0.72338  -1.282 0.199843
hxsmoking.relNo                  -0.20705    1.36935  -0.151 0.879814
n.relN1a                          2.28599    0.87184   2.622 0.008741 **
n.relN1b                          2.54797    0.78142   3.261 0.001111 **
response.relBiochemical Incomplete -3.33817  0.94387  -3.537 0.000405 ***
response.relExcellent            -8.13436    1.27405  -6.385 1.72e-10 ***
response.relIndeterminate        -5.94255    0.97037  -6.124 9.13e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)
```

- Age, N1a, N1b, Biochemical Incomplete, Excellent, and Indeterminate are statistically significant predictors of differentiated thyroid cancer recurrence score at the 5% significance level

```
    Null deviance: 442.657  on 363  degrees of freedom
Residual deviance:  84.742  on 355  degrees of freedom
AIC: 102.74

Number of Fisher Scoring iterations: 8
```

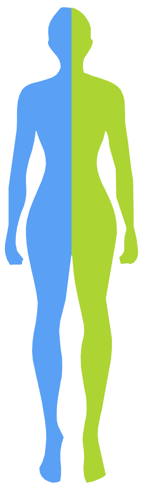
$$P^0(treatment) = \frac{1}{-1.3204 - 0.0486 \cdot Age\ 50 + 0.9274 \cdot Female - 2.5480 \cdot N1b + 5.9425 \cdot Indeterminate\ Response}$$

$$= \frac{1}{2.770904}\ 0.3608931 \approx\ 0.3609 \approx 36.09\ \%$$

```
binary.prediction <- predict(binary.fitted.model, data.frame(age=50, gender.rel="F",
hxsmoking.rel="Yes", n.rel="N1b", response.rel = "Indeterminate"), type="response")
print(binary.prediction)
```

```
        1
0.3604712
```

The predicted probability of treatment for a differentiated thyroid cancer recurrence patient, a 50-year-old woman with a history of smoking, N1b, and an indeterminate response using the Binary Logistic Regression Model with binomial logit link function is 36%.



Probit Model

```
glm(formula = recurred ~ age + gender.rel + hxsmoking.rel + n.rel +
    response.rel, family = binomial(link = "probit"), data = differentiated_thyroid_cancer_recurrence.data)

Coefficients:
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                      0.80461    0.84237   0.955 0.339491
age                              0.02310    0.01063   2.172 0.029828 *
gender.relF                     -0.49739    0.36834  -1.350 0.176902
hxsmoking.relNo                 -0.14012    0.65086  -0.215 0.829546
n.relN1a                         0.96566    0.45397   2.127 0.033408 *
n.relN1b                         1.34384    0.37236   3.609 0.000307 ***
response.relBiochemical Incomplete -1.70501  0.45442  -3.752 0.000175 ***
response.relExcellent           -4.14319    0.53172  -7.792 6.59e-15 ***
response.relIndeterminate       -3.06370    0.42804  -7.158 8.21e-13 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)
```

- At the 5% significance level, age, N1a, N1b, biochemical incomplete, excellent, and indeterminate are significant predictors, as in the Probit model.

```
    Null deviance: 442.66  on 363  degrees of freedom
Residual deviance:  85.82  on 355  degrees of freedom
AIC: 103.82

Number of Fisher Scoring iterations: 9
```

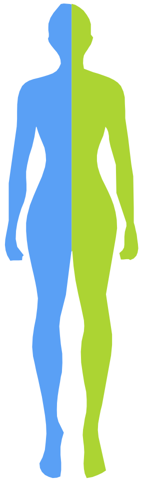
```
probit.prediction <- predict(probit.fitted.model,
data.frame(age=50, gender.rel="F", hxsmoking.rel="Yes",
n.rel="N1b", response.rel = "Indeterminate"), type="response")
print(probit.prediction)
```

```
        1
0.3982731
```

```
probit = (0.8046+0.0231*50-0.4974+1.3438-3.0637)
print(probit)
print(pnorm(probit))
```

```
[1] -0.2577
[1] 0.3983192
```

The predicted probability of treatment for a differentiated thyroid cancer recurrence patient, a 50-year-old woman with a history of smoking, N1b, and an indeterminate response using the Probit Model with binomial probit link function is 39.8%.



Complementary Log-Log Model

```
glm(formula = recurred ~ age + gender.rel + hxsmoking.rel + n.rel +
    response.rel, family = binomial(link = "cloglog"), data = differentiated_thyroid_cancer_recurrence.data)

Coefficients:
                               Estimate Std. Error z value Pr(>|z|)
(Intercept)                     0.60498    0.91210   0.663 0.507146
age                             0.02040    0.01196   1.705 0.088242 .
gender.relF                    -0.98466    0.47062  -2.092 0.036415 *
hxsmoking.relNo                -0.06733    0.64150  -0.105 0.916407
n.relN1a                        0.78447    0.53236   1.474 0.140593
n.relN1b                        1.68642    0.50957   3.309 0.000935 ***
response.relBiochemical Incomplete -1.35855    0.43955  -3.091 0.001996 **
response.relExcellent          -6.00962    1.04324  -5.761 8.38e-09 ***
response.relIndeterminate      -3.49937    0.53919  -6.490 8.58e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)
```

- Age, N1a, N1b, biochemical incomplete, excellent, and indeterminate, is still a significant predictor at the 5% level, since their p-values are smaller than 0.05, as seen in the Binary Logistic Regression and Probit model.

```
    Null deviance: 442.657  on 363  degrees of freedom
Residual deviance:  87.432  on 355  degrees of freedom
AIC: 105.43

Number of Fisher Scoring iterations: 10
```

```
cloglog.prediction <- predict(cloglog.fitted.model, data.frame(age=50,
gender.rel="F", hxsmoking.rel="Yes", n.rel="N1b",
response.rel = "Indeterminate"), type="response")
print(cloglog.prediction)
```

```
        1
0.2661779
```

```
cloglog=1-exp(-exp(0.6050+0.0204*50-0.9847+1.6864-3.4994))
print(cloglog)
```

```
[1] 0.2662083
```

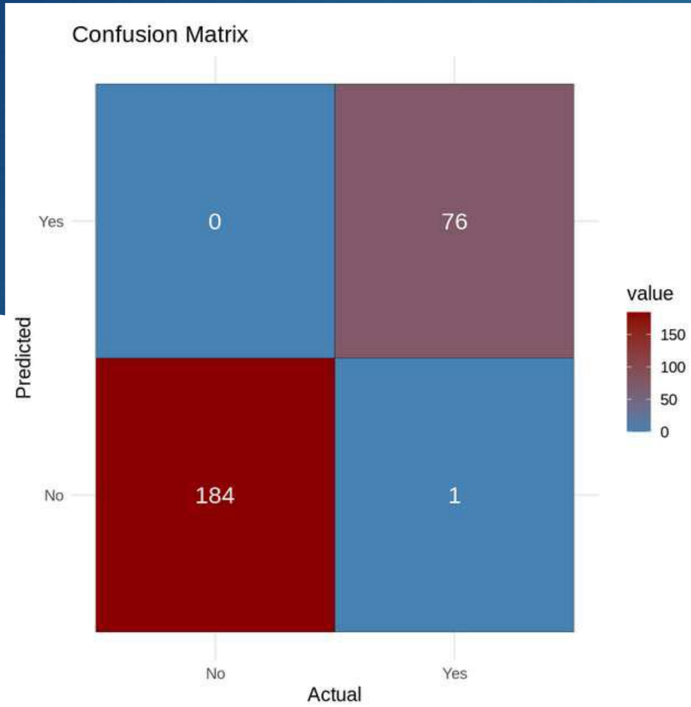
The predicted probability of treatment for a differentiated thyroid cancer recurrence patient, a 50-year-old woman with a history of smoking, N1b, and an indeterminate response using the Complementary Log-Log model with a binomial cloglog link function is 26.6%.

AIC and AICc and BIC Values

- AIC (Akaike Information Criterion), AICc (corrected AIC), and BIC (Bayesian Information Criterion) Values
- The Binary Logistic Regression with 102.7418 for AIC, 103.2503 for AICc, and BIC 137.8162 has the smallest values in all AIC, AICc, and BIC criteria and the best fit.

	Binary Logistic Regression	Probit Model	Complementary Log-Log
AIC	102.7418	103.8201	105.432
AICc	103.2503	104.3286	105.9404
BIC	137.8162	138.8945	140.5064

Random Forest



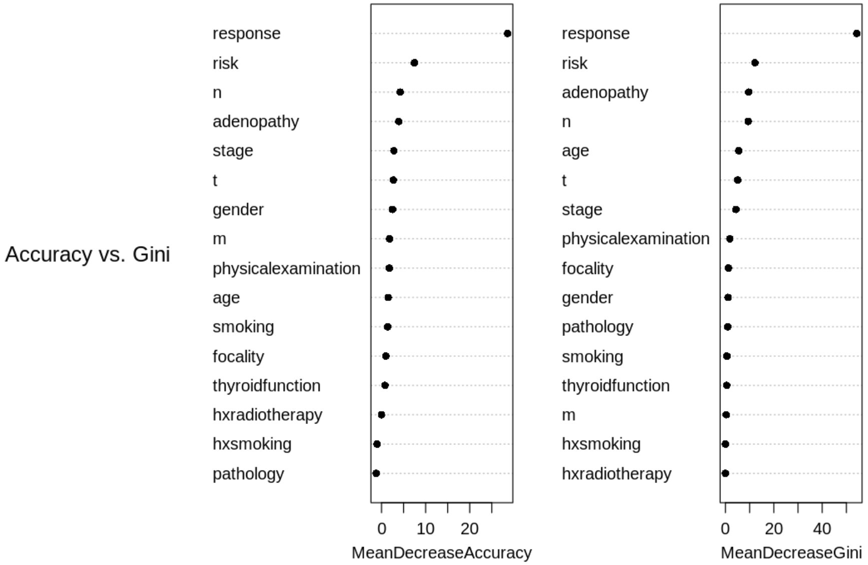

```
   Cell Contents
|-------------------------|
|                       N |
| Chi-square contribution |
|           N / Row Total |
|           N / Col Total |
|         N / Table Total |
|-------------------------|


Total Observations in Table:  261


                                    | train$recurred
diff_thyroid_cancer_recurrence_preds_train |        No |       Yes | Row Total |
-------------------------------------------|-----------|-----------|-----------|
                                       No |       184 |         1 |       185 |
                                          |    22.011 |    52.597 |           |
                                          |     0.995 |     0.005 |     0.709 |
                                          |     1.000 |     0.013 |           |
                                          |     0.705 |     0.004 |           |
-------------------------------------------|-----------|-----------|-----------|
                                      Yes |         0 |        76 |        76 |
                                          |    53.579 |   128.032 |           |
                                          |     0.000 |     1.000 |     0.291 |
                                          |     0.000 |     0.987 |           |
                                          |     0.000 |     0.291 |           |
-------------------------------------------|-----------|-----------|-----------|
                             Column Total |       184 |        77 |       261 |
                                          |     0.705 |     0.295 |           |
-------------------------------------------|-----------|-----------|-----------|
```

True Positive is 76
False Positive is 1
True Negative is 184
False Negative is 0

```
Accuracy:            0.9962
Misclass. Rate:      0.0038
Precision:           0.987
Recall (Sensitivity):  1
Specificity:         0.9946
F1 Score:            0.9935
False Positive Rate (FPR):  0.0054
False Negative Rate (FNR):  0
True Positive Rate (TPR):  0.4108
True Negative Rate (TNR):  2.4211
Negative Predictive Value (NPV):  1
One Minus Specificity:  0.0054
```



CONCLUSION & FUTURE WORK

- Random Forest, performed the best with 99.62% accuracy, while the Complementary Log-Log Model had an accuracy of 26.62% which is the lowest accuracy.

- Utilize Convolution Neural Networks and Artificial Neural Networks to test and train our model with machine learning algorithms.

References

- https://www.cancer.gov/publications/dictionaries/cancer-terms/def/differentiated-thyroid-cancer
- https://together.stjude.org/en-us/conditions/cancers/differentiated-thyroid-cancer.html
- https://archive.ics.uci.edu/dataset/915/differentiated+thyroid+cancer+recurrence
- https://www.routledge.com/Advanced-Regression-Models-with-SAS-and-R/Korosteleva/p/book/9780367732424?srsltid=AfmBOop1t7LOcOlEl6_B0sxq_x9IXVLcwMpuYe08VxsO7aVLGqbqOC5O
- https://www.stat.berkeley.edu/%7Ebreiman/randomforest2001.pdf

THANK YOU

Any Questions?

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